

Sixteen-Frame Interferometer for a Study of a Pinch Dynamics in PF-1000 Device

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Multiframe laser interferometer system for studies of rapidly varying processes in dense, high-temperature plasma in plasma-focus device is described. The setup of sixteen-frame interferometer contains the Mach-Zehnder system and the completely new, innovative beams separator which allows obtaining sixteen interferometric frames in one discharge. Each interferogram is recorded separately with the same linear size and quality. Delay from the first to the last frame is 220 ns. The result obtained using this system is given. It demonstrates its quality and suitability for investigations of dynamics of fast, irreproducible processes in PF-1000.

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1 Introduction

Devices called Plasma-Focus (PF) play an important role among instrumentation used in generation of the high-temperature plasma [1]. The interest in them is associated with a fact that they are also known to be sources of intensive neutrons in laboratory conditions [2]. The research carried out with these devices is focused around the following topics:

- physical phenomena associated with discharge in PF devices;
- theoretical description of physical phenomena taking place during discharge;
- optimization of PF devices for production of neutrons or generation of beams of electrons and ions.

Despite the considerable progress noted in this area of research the understanding of basic processes taking place in the PF devices is still incomplete [3]. Therefore, construction of a new experimental setup based on an interferometer capable of recording up to 16 images will allow for further study of phenomena during one discharge. Intervals 10 ns alternating with 20 ns between frames allow the detailed study of time-resolved phenomenon. The experimental data collected with this apparatus is a unique source of information that makes possible not only to continue the progress in understanding the discharge phenomena but, also let optimize and control these processes.

2 Experimental setup

The setup with an interferometer, described in detail below, made possible acquisition of sixteen interferometric images during one discharge in the PF-1000 device.

A schematic diagram of this system is given in Fig. 1. The main element of the diagnostic setup is the Mach-Zehnder interferometer shown in Fig. 2. In the studies of the PF phenomena this apparatus used in the experiment has advantages over other types of interferometers for the following reasons:

- sufficiently large field of vision of the interferometer;

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- possibility to follow in time the fast changeable plasma fields due to a single passage of a laser light beam through the plasma for obtaining the interferometric image.

The most important element of the interferometric setup is the light source. In our case it is the Nd:YLF laser. This laser emits light with the base wavelength of $1.06\ \mu\text{m}$ and process second, third and fourth harmonic frequencies. For interferometric purposes we were using the second harmonic with the wavelength of $527\ \text{nm}$, i.e. the visible green light. The energy carried by this wavelength in a single laser pulse is approximately $500\ \text{mJ}$, with energy stability of 7% , time duration less than $1\ \text{ns}$, the diameter of the light beam $12\ \text{mm}$, and the time jitter of the generated laser pulses $\pm 1\ \text{ns}$. These parameters play a critical role in the studies especially to precise establish time correlation of all phenomena taking place during plasma discharge.

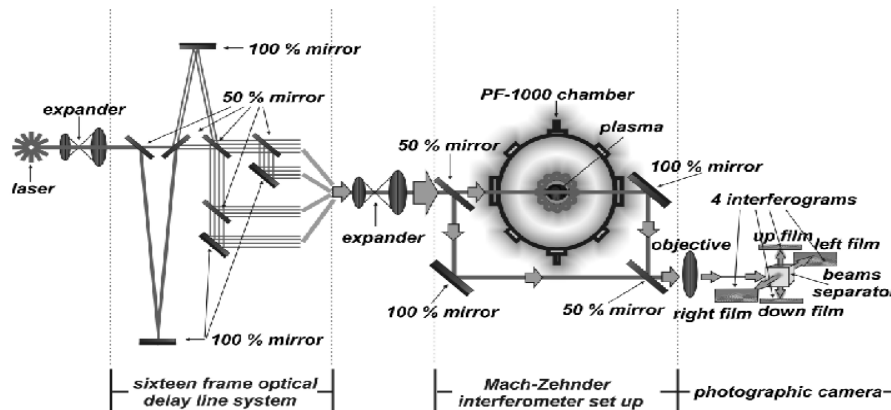


Fig. 1 A schematic diagram of sixteen frame interferometer system.

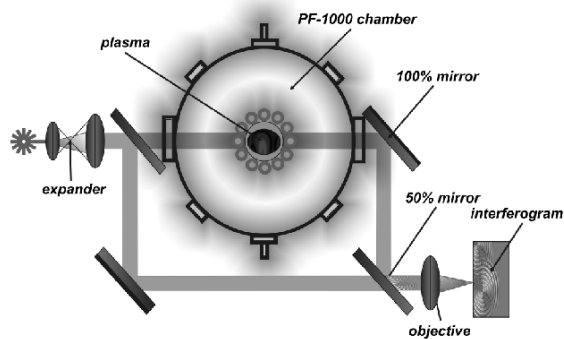


Fig. 2 Mach-Zehnder interferometer setup for plasma density measurements.

An optical telescope increasing the beam diameter up to $60\ \text{mm}$ was installed in front of the laser. Introduction of the telescope in such a location forms a parallel light beam and cancels the natural divergence of the laser beam which is very undesirable beam parameter in the multiframe interferometry especially when operating with optical path dozens of meters in length.

An optical delay line shown in Fig. 3 allows for spatial selection between frames delays in the range from $0\ \text{ns}$ to $220\ \text{ns}$ and ensures the same linear sizes of interferograms in each frame. Two types of optical elements were used in this setup: 100% mirrors permits and $50/50$ beam splitters. They direct light beams corresponding to each frame at a small angle to the optical axis of the setup. In such a way we obtain sixteen light beams in form of 4×4 matrix around the center of optical axis of first Mach-Zehnder mirror. Such configuration allows obtaining fully separate in space frames with the same linear sizes and quality during their registration (Fig. 4).

The laser beams passing through the optical delay line enters into expander which increases beam diameter to the size of $150\ \text{mm}$. Collimated beams are directed to the mirrors of the Mach-Zehnder interferometer. The active diameter of the mirrors in the interferometer is $200\ \text{mm}$ so it capable of complete observation of the phases of the possible PF phenomena.

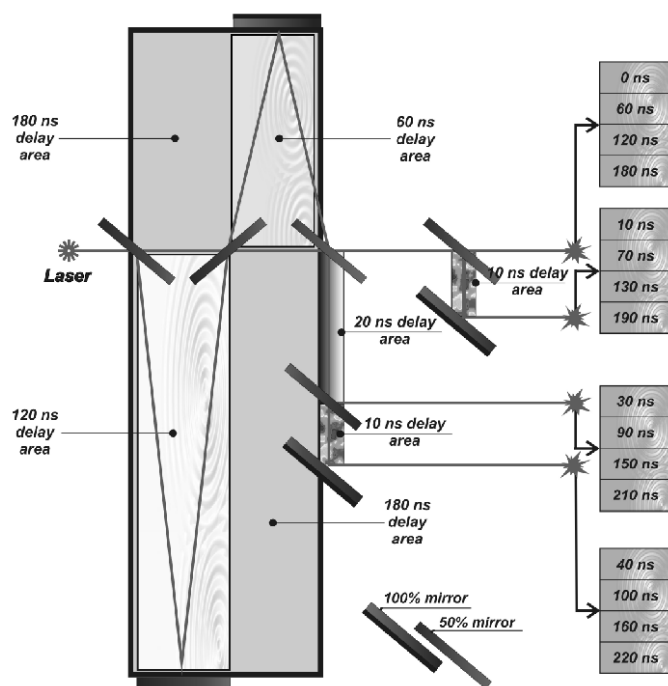


Fig. 3 Sixteen frame optical delay line diagram.

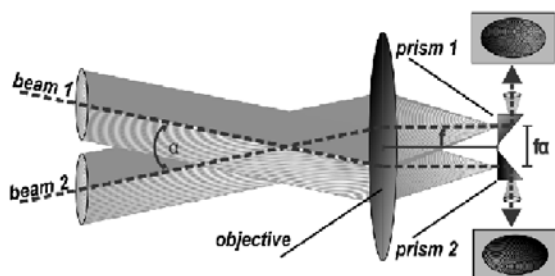


Fig. 4 Idea of beams separation.

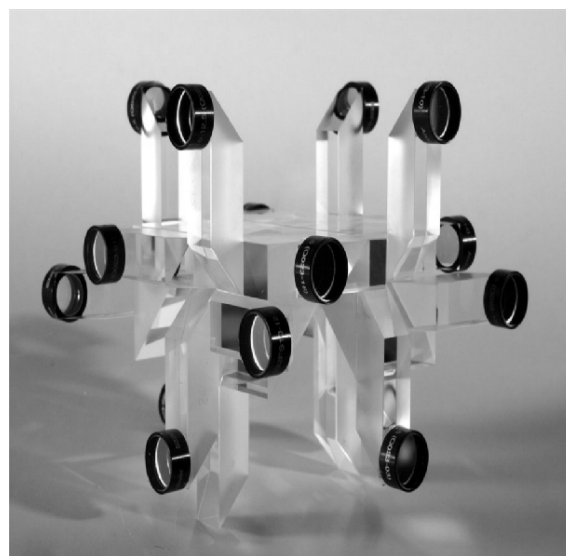


Fig. 5 Beams separator.

A recording system has been built for the specific needs of this experiment and it allows for registration of interferometric images on photographic plates, and provides spatial separation of frames into four planes of registration. A special setup of prisms, specially constructed for the needs of this experiment (Fig. 5) distributes 16 delayed in time beams onto four photographic plates of four images each. Beam outgoing of each of the sixteen prisms is passing through the system of optical interferometric filters preventing huge plasma irradiance in visible spectrum. The camera is equipped with optical filters in front, and four 130×180 mm photographic cassettes mounted at the exits, opened during exposure. For registration of images a special type of film was selected for optimal contrast and resolution of the registered images.

3 Results

Below we present 16 interferograms (Fig. 6). These experimental results were obtained for one discharge in PF-1000 device. Vacuum chamber initially was under pressure of 2×10^{-5} hPa, filled with deuterium gas under pressure of 2,6 hPa. Total neutron yield in this shot was $1,3 \times 10^{11}$.

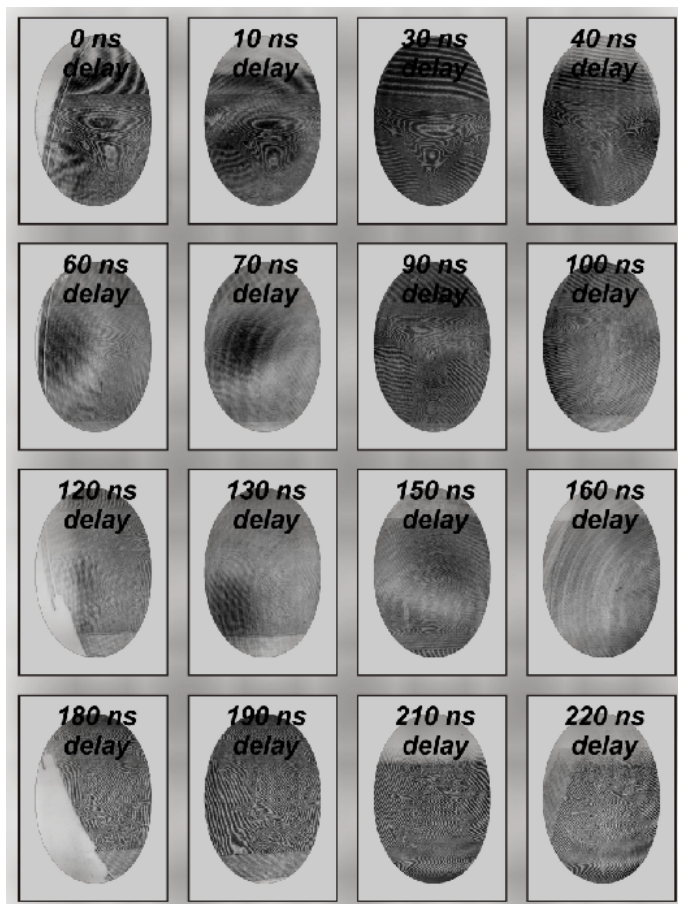


Fig. 6 Sixteen interferograms from one shot from PF-1000.

4 Conclusions

Light interferometry is an extremely valuable research method. The ability to quantitatively determine plasma density distribution from time developed interferograms is giving the information about:

- space-time evolution of the electrons density distribution in the inspected configuration;
- efficiency of the gas gathering by the plasma sheath;
- degree of compression of plasma;
- the total plasma volume in the pinch area;
- plasma leakage from pinch area.

Light interferometry has been using for many years and this technique is still in progress. Several solutions had been developed [4, 5, 6, and 7]:

- use of several lasers;

- use beam splitter and mirror combinations;
- beam splitter, mirror and half-wave plate combinations.

All interferometric configurations are difficult to align especially when we try to obtain multiple frames in one discharge. The main problem in a multiple-frame technique was that when was used an image-plan configuration to obtain high quality interferograms the minimum separation angle became too large. We resolved this problem by inventing the new beams separator which is making possible to record 16 interferograms during one discharge.

The sixteen-frame interferometry provides high resolution information ordered in time. It allows studying all phenomena in hot plasma [8] and gives information needed to verify the models describing plasma propagation and the structure of plasma coat.

Using our setup it is possible to:

- reconstruct evolution of pinch geometry;
- determine linear density of particles;
- determine distribution of plasma density in the pinch column;
- correlate in time all this phenomena with evolution of X-ray and neutron production;
- characterize the MHD instabilities in the processes of plasma development;
- connect speed of development of the MHD $m=0$ instability of the plasma string with efficiency of neutron production;
- analyze connections between processes of rising and decay of the plasma string with neutrons efficiency;
- determine the plasma density during the PF phenomena development;
- precise correlate PF phenomena development with other diagnostics data which gives information about dependence between speed of thermonuclear synthesis reaction and the parameters of the PF setup generator.

All these information let determine law of scaling for PF setups, as well as provide valuable reference data for various work regimes of these devices.

References

- [1] L. Soto, "New Trends and Future Perspectives on Plasma Focus Research", *Plasma Physics and Controlled Fusion* **47**, A361-A381 (2005).
- [2] H. Schmidt, P. Kubes, M.J. Sadowski, M. Scholz, "Neutron Emission Characteristics of Pinched Dense Magnetized Plasma", *IEEE Trans. Plasma Science* **34**, 2363-2367 (2006).
- [3] V.I. Krauz, "Progress in Plasma Focus Research and Applications", *Plasma Physics and Controlled Fusion* **48**, B221-B229 (2006).
- [4] S. Denus, A. Kasperczuk, M. Paduch, M. Pokora, Z. Wereszczyński, "Multi-frame interferometric setup for studying quick-change processes", *Journal of Technical Physics* **18**, 395-405 (1977).
- [5] A. Kasperczuk, J. Makowski, M. Paduch, K. Tomaszewski, Z. Wereszczyński, J. Wołowski, J. Wyzgał, "Application of Fast Three-frame Interferometry to Investigation of Laser Plasma", *Proc. SPIE* **22**, 449 (1994).
- [6] L. Soto, H. Chuaqui, M. Skowronek, "Eight Frame Holographic Interferometry System for Transient Plasma Diagnostics", *Applied Optics* **34**, 7831 (1995).
- [7] G.V. Ostrovskaya, "Holographic Diagnostics of the Plasma: A Review", *Technical Physics*, **53**, 1103-1128 (2008).
- [8] P. Kubes, M. Paduch, T. Pisarczyk, M. Scholz, T. Chodukowski, D. Klir, J. Kravarik, K. Rezac, I. Ivanova-Stanik, L. Karpinski, K. Tomaszewski, E. Zielinska, "Interferometric Study of Pinch Phase in Plasma Focus Discharge at the Time of Neutron Production", *IEEE Transactions on Plasma Science* **37**, 2191-2196 (2009).